

SHETH L.U.J AND SIR MV COLLEGE
PRACTICAL NO:- 8

AIM: Implementation of DSR Routing Protocol

```
mvl@IT-09: ~$ ...
(file "dsr_sim.tcl" line 1)
mvl@IT-09: ~$ nano dsr_sim.tcl
mvl@IT-09: ~$ cat dsr_sim.tcl
set ns [new Simulator]
$ns node-config -adhocRouting DSR \
-llType LL \
-macType Mac/802.11 \
-ifqType Queue/DropTail/PriQueue \
-ifqLen 50 \
-antType Antenna/OmniAntenna \
-propType Propagation/TwoRayGround \
-phyType Phy/WirelessPhy \
-channelType Channel/WirelessChannel \
-topoInstance $topo \
-agentTrace ON \
-routerTrace ON \
-macTrace OFF
mvl@IT-09: ~$ nano dsr_sim.tcl
mvl@IT-09: ~$ ls
aodv_out.nam  aodv_sim.tcl  dsr.tr        dsr_sim.tcl    nam_1.15-6.dsc  nam_error.txt
aodv_out.tr   dsr.nam       dsr_practical  nam_1.15-6.debian.tar.xz  nam_1.15.orig.tar.gz  'nano aodv_sim.tcl'
mvl@IT-09: ~$ nam dsr_out.nam
nam:
mvl@IT-09: ~$ nam dsr.nam
nam:
mvl@IT-09: ~$ wc -l dsr.nam
0 dsr.nam
mvl@IT-09: ~$ head -20 dsr.nam
mvl@IT-09: ~$ nano dsr_sim.tcl
mvl@IT-09: ~$ rm -f dsr.nam dsr.tr
mvl@IT-09: ~$ ns dsr_sim.tcl
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
can't read "phyType_": no such variable
while executing
"$node add-interface $chan $propInstance_ $llType_ $macType_ $ifqType_ $ifqLen_ $phyType_ $antType_ $topoInstance_ $inerrProc_ $outerrProc_ $FECProc_"
(procedure "_o3" line 83)
(Simulator create-wireless-node line 83)
invoked from within
"_o3 create-wireless-node"
("eval" body line 1)
invoked from within
```

```
GNU nano 7.2 dsr_sim.tcl
set ns [new Simulator]

set tracefd [open dsr.tr w]
set namfd [open dsr.nam w]

set topo [new Topography]
$topo load_flatgrid 500 500

set god_ [create-god 3]

$ns trace-all $tracefd
$ns namtrace-all-wireless $namfd 500 500

set chan_1_ [new Channel/WirelessChannel]

$ns node-config \
-adhocRouting DSR \
-llType LL \
-macType Mac/802.11 \
-ifqType Queue/DropTail/PriQueue \
-ifqLen 50 \
-antType Antenna/OmniAntenna \
-propType Propagation/TwoRayGround \
-phyType Phy/WirelessPhy \
-channel $chan_1_ \
-topoInstance $topo \
-agentTrace ON \
-routerTrace ON \
-macTrace OFF

set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]

$n0 random-motion 0
$n1 random-motion 0
$n2 random-motion 0
```

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```
GNU nano 7.2 dsr_sim.tcl
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]

$ns random-motion 0
$ns random-motion 0
$ns random-motion 0

$ns initial_node_pos $n0 50
$ns initial_node_pos $n1 150
$ns initial_node_pos $n2 250

set udp [new Agent/UDP]
$ns attach-agent $n0 $udp

set null [new Agent/Null]
$ns attach-agent $n2 $null

$ns connect $udp $null

set cbr [new Application/Traffic/CBR]
$cbr set packetSize 512
$cbr set interval 0.1
$cbr attach-agent $udp

$ns at 1.0 "$cbr start"
$ns at 8.0 "$cbr stop"

proc finish {} {
    global ns tracefd namfd
    $ns flush-trace
    close $tracefd
    close $namfd
    exit 0
}

$ns at 10.0 "finish"
```

```
invoked from within
"God instance"
(procedure ".o13" line 113)
(Object next line 113)
invoked from within
".o13 next .o11 .o12 LL Mac/802_11 Queue/DropTail/PriQueue 50 Phy/WirelessPhy Antenna/OmniAntenna .o10 {} {}"
(eval body line 1)
invoked from within
eval $self next $args
(procedure ".o13" line 2)
(SRNodeNew add-interface line 2)
invoked from within
$node add-interface $chan $propInstance_ $llType_ $macType_ $ifqType_ $ifqlen_ $phyType_ $antType_ $topoInstance_ $inerrProc_ $outerrProc_ $fecProc_
(procedure ".o3" line 83)
(Simulator create-wireless-node line 83)
invoked from within
".o3 create-wireless-node"
(eval body line 1)
invoked from within
eval $self create-wireless-node $args
(procedure ".o3" line 23)
(Simulator node line 23)
invoked from within
$ns node
invoked from within
set n0 [$ns node]
(file "dsr_sim.tcl" line 29)
mvlu@IT-09:~$ nano dsr_sim.tcl
mvlu@IT-09:~$ rm -f dsr.nam dsr.tr
ns dsr_sim.tcl
num_nodes is set 3
INITIALIZE THE LIST xListHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
mvlu@IT-09:~$ wc -l dsr.nam
510 dsr.nam
mvlu@IT-09:~$ nam dsr.nam
nam:
mvlu@IT-09:~$ which nam
/usr/bin/nam
mvlu@IT-09:~$ nam -v
```